

Web3BlockSet: A Dataset for Empirical Research in Blockchain-Oriented Software Engineering

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MSR '26 · Data and Tool Showcase Track · April 13–14, 2026 · Rio de Janeiro, Brazil

INTRODUCTION

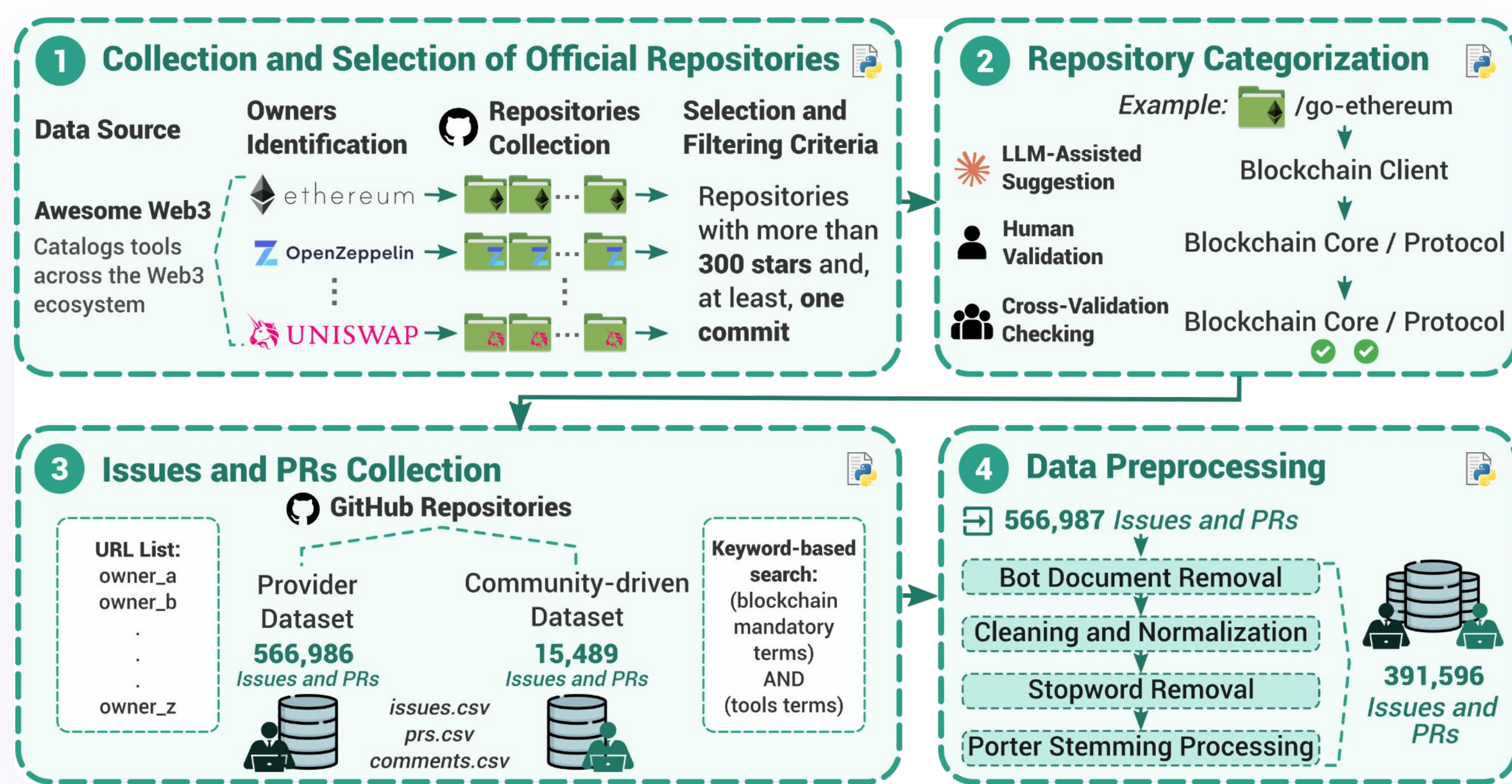
Blockchain-Oriented Software Engineering (BOSE) (Porru et al., 2017) establishes a dedicated field for decentralized systems development, where immutability, peer-to-peer networks, and security mechanisms require distinct practices. Prior work (Das et al., 2022; Opu et al., 2025) examines blockchain repositories but neither separates technology providers from adopters nor categorizes projects by blockchain stack layers, limiting investigations into how challenges propagate across the ecosystem. **Web3BlockSet** addresses this through a Mining Software Repositories approach on GitHub, supporting empirical investigations in BOSE across different aspects and categories.

Our Contribution:

- (1) Provider-to-adopter coverage, from platform maintainers to community developers
- (2) Repository-level categorization across blockchain stack layers (protocols, SDKs, DeFi, wallets, tools)
- (3) Rich metadata with 20+ fields capturing repository characteristics, user interactions, and temporal evolution

METHODOLOGY

Our methodology follows a Mining Software Repositories (MSR) approach structured in four phases, as shown in **Figure 1**.



Step 1 — Collection & Selection of Official Repositories

Used Awesome Web3 (763★, 165 forks, 83 contributors) to identify 80 organizations. Filtered repositories with more than 300 stars and at least one commit, resulting in 381 official repositories.

Step 2 — Repository Categorization

Hybrid approach: (a) LLM-assisted suggestion using Claude (claude-3-5-sonnet); (b) Human validation by an expert with 5+ years in BOSE; (c) Cross-validation by 4 independent BOSE researchers → 92.4% overall agreement.

Step 3 — Issues & Pull Requests Collection

Provider Dataset: all Issues/PRs from 381 official repositories (383,387 documents). Community Dataset: keyword-based search combining mandatory blockchain terms ("blockchain", "web3", "smart contract") with tool and library names → 8,209 documents from 4,500 repositories.

Step 4 — Data Preprocessing

Bot filtering, Markdown removal, Unicode normalization (NFKD), ASCII conversion, stopword removal (NLTK + custom developer terms), Porter stemming. Documents with fewer than 15 words after preprocessing were discarded.

CHARACTERIZING THE WEB3BLOCKSET

Official Repository Overview

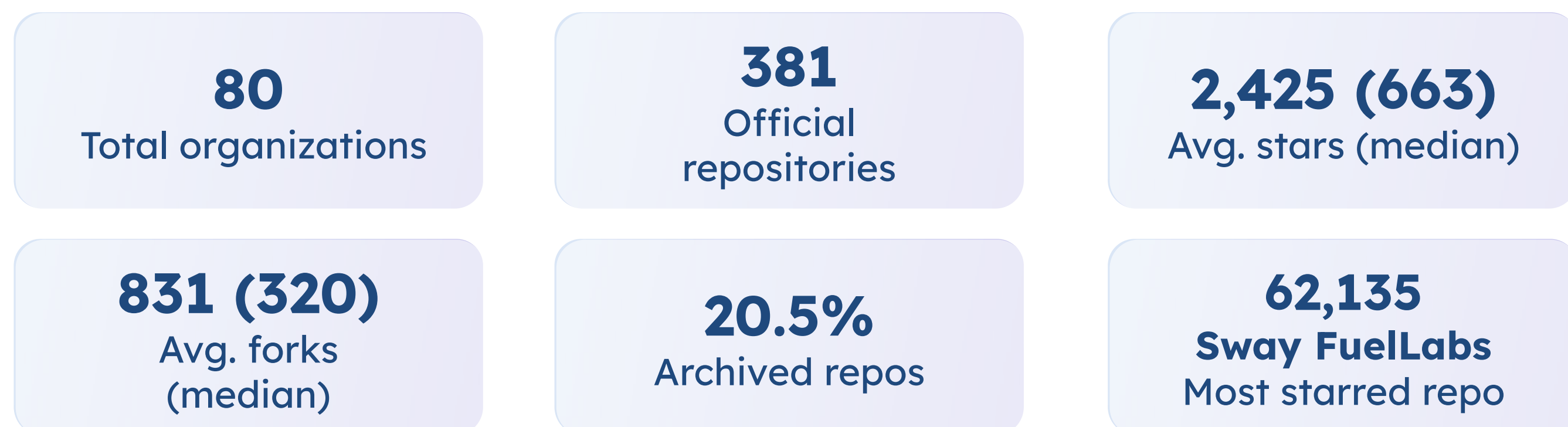
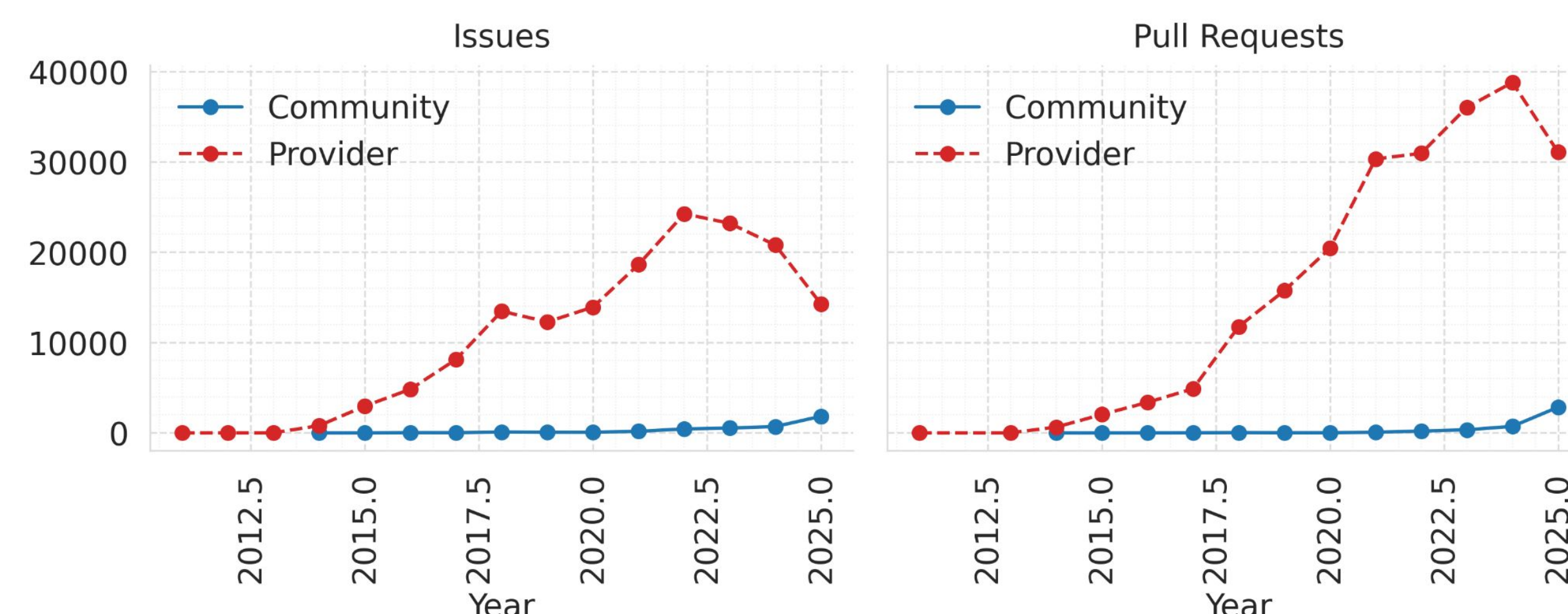


Figure 2 shows the distribution of repository categories for provider organizations.

Web3BlockSet captures developer interactions through Issues and Pull Requests across both datasets). **Figure 3** shows the temporal distribution of Issues and PRs created over the years.



Web3BlockSet Schema and Data Structure

File	Fields
repositories.csv (381 records)	organization, description, stars, forks, language, created_at, updated_at, archived, size_kb, has_issues, open_issues, default_branch, clone_url, html_url, repository_category
issues_prs.csv (391,596 records)	repository, owner, issue_id, issue_number, issue_title, issue_body, state, created_at, updated_at, closed_at, author, author_id, locked, comments_count, commits_count, labels, type, concatenated_comments, comment_authors, comments_count_filtered, year, raw_text, clean_text, stemmed_text, data_source, repository_category

USE CASE: BLOCKCHAIN DEX ANALYSIS

Web3BlockSet was applied in an ongoing study on Blockchain Developer Experience (BcDEX) in the context of Blockchain as a Service (BaaS) platforms. The study analyzed more than 12,000 Issues and PRs from provider and community repositories using BERTopic for topic modeling and the SPACE framework for productivity analysis across five dimensions: Satisfaction and Well-Being, Performance, Activity, Communication and Collaboration, and Efficiency and Flow. Results showed complementary specialization:

- **Provider Repositories:** Infrastructure abstraction (wallet integration, network RPC)
- **Community Repositories:** Implementation challenges (smart contract development, network configuration)

FUTURE RESEARCH QUESTIONS

RQ1 — Technology Adoption & Migration: Which platforms are adopted by community developers, and how do adoption patterns change over time? What factors predict blockchain tool abandonment, and how long do projects remain maintained?

RQ2 — Cross-Platform Development Challenges: Do different blockchain platforms exhibit distinct developer challenges and issue types? How does programming language choice correlate with issue resolution time and developer sentiment?

RQ3 — Collaboration & Contribution Patterns: How do collaboration networks differ between core maintainers and community contributors? What factors predict long-term contributor retention in blockchain projects?

RQ4 — Issue Resolution & Maintainability: How does issue resolution time vary across repository categories and organizational size? Does repository popularity correlate with maintainer responsiveness and contribution acceptance rates?

RQ5 — Security Vulnerability Disclosure: How do vulnerability disclosure and resolution practices differ across blockchain platforms? How do security vulnerabilities in provider repositories propagate to community projects?

RQ6 — Evolution of Development Terminology: How does blockchain development terminology evolve over time, and which terms predict emerging technologies or practices?

CONCLUSIONS

Limitations: Provider dataset relies on the Awesome Web3 list, may not cover all relevant blockchain projects. Community dataset (8,209 documents) is substantially smaller than provider dataset (383,387 documents), which may limit certain comparative analyses. Validation was based on category agreement rather than independent categorization by all reviewers.

Future Works: Incorporate additional repository discovery methods beyond Awesome lists and keyword search. Add sentiment analysis scores, code metrics, and dependency network data. Quarterly dataset updates to maintain temporal relevance and capture ecosystem evolution.

Funding: CNPq-Brazil, Universal grant 404406/2023-8 ·

CAPES — Funding Code 001.

Repository



Paper



Key References

- Porru, S., Pinna, A., Marchesi, M., & Tonelli, R. (2017). Blockchain-oriented software engineering: challenges and new directions. ICSE-C '17.
- Das, A., Uddin, G., & Ruhe, G. (2022). An empirical study of blockchain repositories in GitHub. EASE '22.
- Opu, M. N. I., Islam, M. S., Rouhani, S., & Chowdhury, S. (2025). Understanding the Issue Types in Open Source Blockchain-based Software Projects with the Transformer-based BERTopic. arXiv:2506.11451.
- Forsgren, N., Storey, M.-A., Maddila, C., Zimmermann, T., Houck, B., & Butler, J. (2021). The SPACE of Developer Productivity: There's more to it than you think. ACM Queue, 19(1), 20–48.